

Question I: Chameleonic cobalt

Information was always regarded as the most valuable product resulting from mankind activity. It is not striking that recognition of this fact was followed by numerous efforts aimed at information safety. Cryptography seemed to be a convenient way to reach such safety from unrecorded time. Cryptography cannot be detached from sympathetic ink that becomes visible only after special treatment, for instance, heating. History knows a number of recipes of such ink, among them that based on salts of cobalt(II). Being pale-pink in color, cobalt ink is virtually invisible when dried on paper. However, once heated with a candle flame, a letter written with such ink reveals hidden text colored in bright-blue.

We know other applications of cobalt(II) salts, less secret, but dependent on the color transition described above. Blue granules of silica-gel doped with Co(II) salt and placed into a desiccators to dry some product, become pink at last. This is the signal to regenerate silica-gel (just to dry, since it accumulates too much water). Similarly, a paper soaked with saturated solution of CoCl_2 turns blue in dry air due to formation of $\text{CoCl}_2 \cdot 4\text{H}_2\text{O}$, and changes its color back to pink $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ in a humid environment. Apparently, the paper works as a humidity meter, hygrometer.

1.1. Using the thermodynamic data below, determine the threshold of air humidity (in %) specific to the response of such a hygrometer.

Compound	$-\Delta_f H_{298}^\circ$, kJ mol ⁻¹	S_{298}° , J mol ⁻¹ K ⁻¹
$\text{CoCl}_2 \cdot 6\text{H}_2\text{O}_{(\text{cr})}$	2113.0	346.0
$\text{CoCl}_2 \cdot 4\text{H}_2\text{O}_{(\text{cr})}$	1538.6	211.4
$\text{H}_2\text{O}_{(\text{lq})}$	285.8	70.1
$\text{H}_2\text{O}_{(\text{g})}$	241.8	188.7

The “pink (sometimes, violet) $\leftrightarrow$ blue” color transition described above is related to the reconstruction of the coordination sphere of Co^{2+} ion: octahedron $\leftrightarrow$ tetrahedron. The examples discussed in a previous section deal with the transition $[\text{Co}(\text{H}_2\text{O})_6]_{\text{oct}}^{2+} \leftrightarrow [\text{Co}(\text{H}_2\text{O})_4]_{\text{tet}}^{2+}$. As a rule, coordination compounds with tetrahedral geometry are less abundant compared to octahedral ones. However, in particular case of Co^{2+} tetrahedral complexes competes with octahedral compounds.

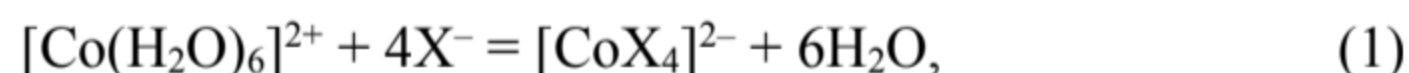
1.2. To understand the reason behind such behavior, consider the following octahedral and tetrahedral complexes:

a) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Cr}(\text{H}_2\text{O})_4]^{3+}$,

b) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Co}(\text{H}_2\text{O})_4]^{2+}$.

Draw diagrams for the case of an octahedral and a tetrahedral ligand field showing clearly the energy levels of all metal 3*d*-orbitals; indicate the *d*-orbital splitting parameter Δ . For each of the ions above use the appropriate diagram and fill it in with the electrons available in the metal *d*-subshell. Calculate the Crystal Field Stabilization Energy (CFSE) for each of the ions. Compare the results and draw a conclusion.

1.3. The following reaction



where $\text{X}^- = \text{Cl}^-, \text{Br}^-, \text{I}^-, \text{SCN}^-$, is used in some textbooks to illustrate Le Chatelier's principle related to equilibrium shifting. If one adds an excess of salt containing X^- , the solution becomes blue, and under dilution with water it turns back pale-pink.

a) Predict the signs of the enthalpy ($\Delta_r H_{298}^\circ$) and entropy ($\Delta_r S_{298}^\circ$) changes for the reaction (1).

b) What effect does temperature produce on the equilibrium (1)?

c) Consider reaction (1) and KCl and KSCN as a source of ions X^- for it. Which salt present in the same molar concentration shifts the equilibrium (1) to the right in a greater extent? Explain using the principle of Hard and Soft Acids and Bases (HSAB).

1.4. Consider a similar equilibrium (2):



a) If $\text{L} = \text{pyridine (py)}$, which ligand X (Cl^- or I^-) helps better shift the equilibrium (2) to the right? Explain using the principle of Hard and Soft Acids and Bases (HSAB).

b) If $\text{L} = \text{PH}_3$, which ligand X (Cl^- or I^-) helps better shift the equilibrium (2) to the right? Explain using the HSAB principle.

c) The coordination compound with the formula $[\text{CoX}_2\text{L}_2]$, where $\text{L} = \text{py}$, $\text{X} = \text{Cl}^-$ exists in two forms colored blue and violet. The structure of the former is quite apparent, whereas that of the latter is less obvious. For the violet form, draw a fragment of its structure large enough to show clearly the coordination mode of the cobalt ion.

With some knowledge of coordination chemistry of Co(II) described above, you may be able to account for the transformations described below.

NaOH solution is added dropwise to a solution of Co(II) under cooling (0 °C), which results in a precipitate of blue color. If the precipitate is left at room temperature (25 °C) for a while, it becomes pink. If an excess of alkali is further added to the precipitate, it dissolves giving blue solution.

1.5. Write down equations corresponding to the transformations described above.

Question II: Ionic equilibria

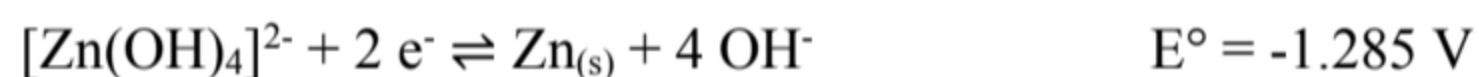
A. Some information about zinc hydroxide

Zinc(II)-hydroxide has a solubility product of $K_{SP}=1.80 \cdot 10^{-17}$.

2.1. Calculate the solubility of zinc(II)-hydroxide in water.

2.2. Calculate the pH-value of a saturated solution of zinc(II)-hydroxide, not taking into account the possible formation of a complex.

The standard potentials of the following reactions are:



2.3. Calculate the complex formation constant for tetrahydroxozincate(II) from Zn^{2+} and OH^- .

2.4. Calculate the solubility of zinc(II)-hydroxide at a pH of 9.58 not taking into account the possible formation of tetrahydroxozincate(II).

2.5. Now calculate the solubility of zinc(II)-hydroxide at a pH of 9.58 taking into account the possible formation of tetrahydroxozincate(II).

2.6. Compare the results of 2.4., and 2.5., and comment using two sentences at most.

B. Borderline cases for buffer equations

The Henderson-Hasselbalch-equation is often used to calculate the pH of conjugate buffer solutions. In deriving this equation the main supposition is, that the buffer acid as well as the buffer base are sufficiently weak so that simplifications are possible. In the following task we will investigate in which cases it is possible to apply the equation and in which not

Dichloro acetic acid has a pK_A -value of 1.290.

2.7. Applying the Henderson-Hasselbalch-equation, calculate the pH of a mixture of dichloro acetic acid/sodium dichloro acetate with the respective concentrations of 0.100 mol/L.

Subsequently, let us check the result of 2.7. Imagine a solution (volume = 1.00 L) of dichloro acetic acid with a concentration of 0.100 mol/L.

2.8. Calculate the pH of this solution (you may neglect the auto protolysis of water, and assume that all activity coefficients are 1). Also calculate the concentrations of all ion species in this solution. Now, 0.1 mol of solid sodium dichloro acetate are added. This causes a corresponding shift of the equilibrium.

2.9. Calculate the pH of the new solution applying the same assumptions as above. What is the difference (in %) of the mathematical solution following Henderson-Hasselbalch compared to the exact value of the H_3O^+ -concentration?

Question III: Platinum Complexes as Anticancer Drugs

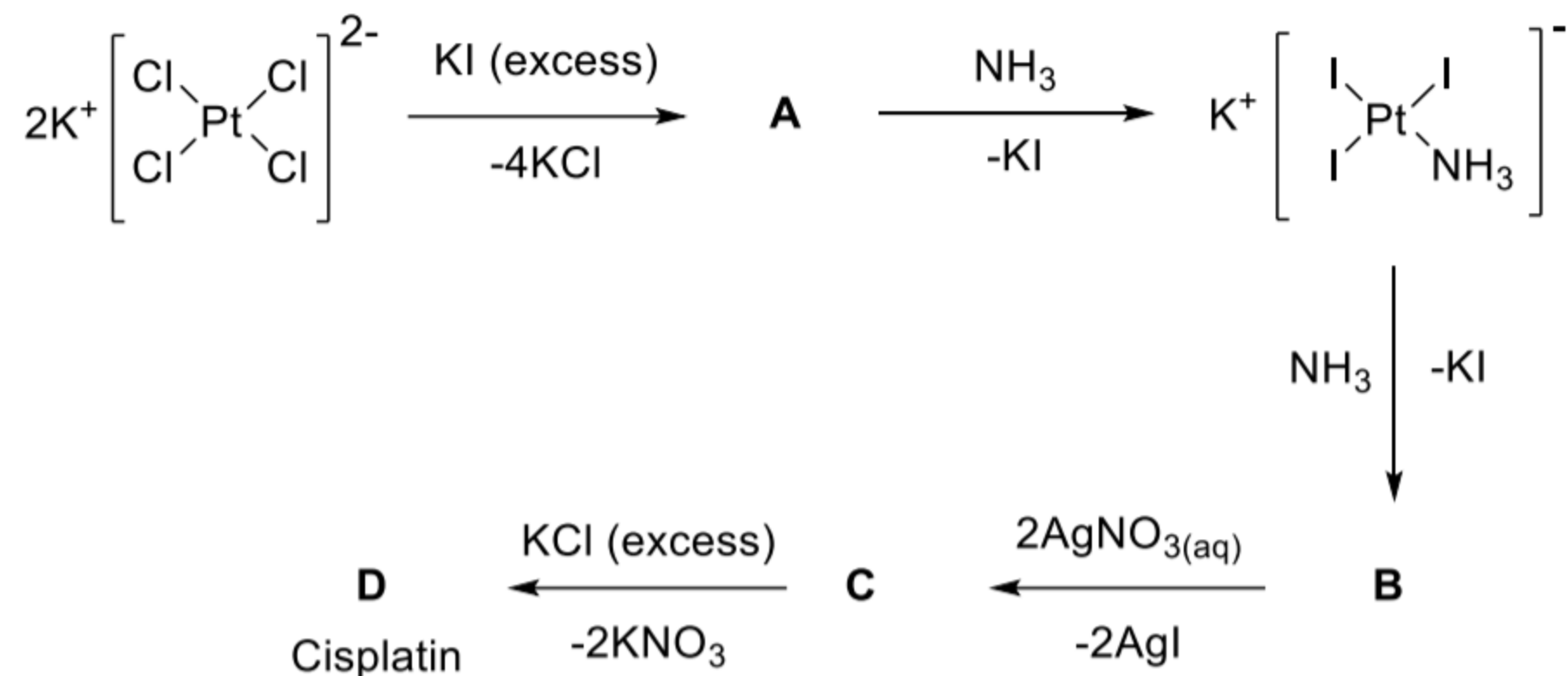
Medicinal inorganic chemistry based on metal-based drugs is broadly defined as the area of research related to metal ions and metal complexes and their clinical applications. It is a new research area that developed from the discovery of the anticancer agent cisplatin. Cisplatin, *cis*-diamminedichloroplatinum(II), is a yellow powder and an anticancer drug widely used in the treatment of a variety of tumors, especially those of the testes, ovaries, head, and neck.

The synthesis of cisplatin starts with $\text{K}_2[\text{PtCl}_4]$, but has undergone several improvements since it was published more than 100 years ago. The main problem is the occurrence of impurities and the formation of the by-product *trans*-platin. Nowadays, the synthetic routes are mostly based on a method published in the 1970s by Dhara. In the initial step, $\text{K}_2[\text{PtCl}_4]$ is reacted with excess KI, and the platinum complex is converted into the iodo analogue (**A**). Subsequently, NH_3 is added to the compound **A** and compound **B** is formed by ligand exchange in which two NH_3 ligands are exchanged with two iodo ligands. **B** is a yellow solid that is filtered, dried, and mixed with the aqueous solution of AgNO_3 . The insoluble AgI can be filtered off and *cis*-diamminediaquaplatinum(II) nitrate (**C**) is formed; then excess KCl is added to the solution of **C** to yield cisplatin (**D**).

The success of the synthesis relies on the strong *trans* effect of the iodo ligands. The spectator ligands *T* that are *trans* to the leaving group in square-planar complexes influence the rate of substitution. This phenomenon is called the *trans* effect. Key point is that a strong σ -donor ligand or π -acceptor ligand greatly accelerates substitution of a ligand situating in the *trans* position. *Trans* effects follow the order given below.

For a *T* σ -donor: $\text{OH}^- < \text{NH}_3 < \text{Cl}^- < \text{Br}^- < \text{CN}^-$, $\text{CH}_3^- < \text{I}^- < \text{SCN}^- < \text{PR}_3$, H^-

For a *T* π -acceptor: $\text{Br}^- < \text{I}^- < \text{NCS}^- < \text{NO}_2^- < \text{CN}^- < \text{CO}$, C_2H_4



3.1. **Write** the formulas of A–D.

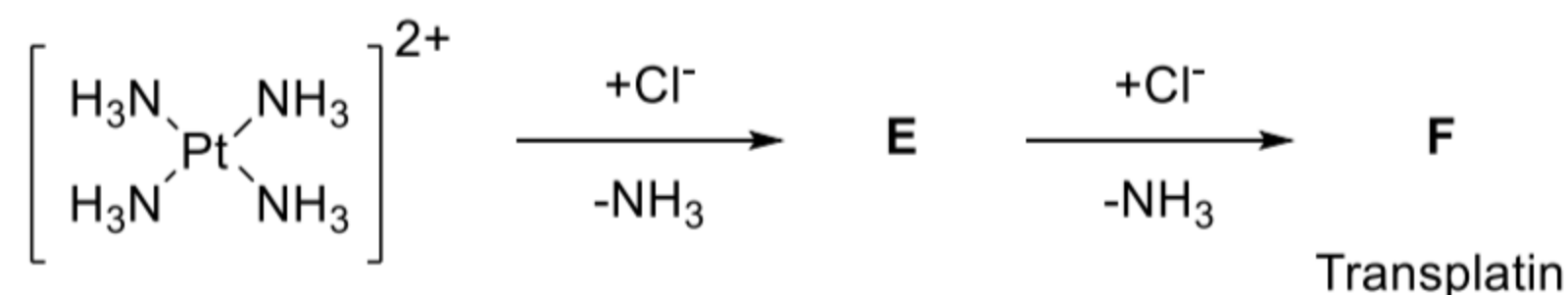
3.2. **Draw** molecular structures of A–D.

3.3. Is the complex **D** polar?

3.4. **Sketch** the d-orbital splitting of cisplatin complex **D** in view of Crystal Field Theory and show the electron distribution diagram.

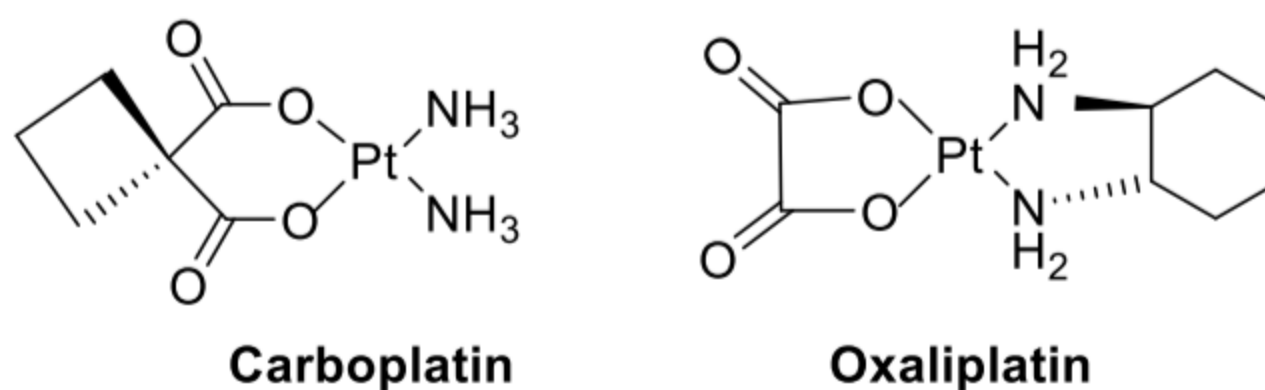
3.5. **Determine** magnetic nature of complex **A**.

The platinum complex binds to DNA and causes cross-linking, which triggers the programmed cell death (apoptosis). However, the other geometrical isomer of the square planar structure transplatin, *trans*-diamminedichloroplatinum(II) (**F**), is not effective for the treatment of cancer. Transplatin is synthesized starting from $[\text{Pt}(\text{NH}_3)_4]^{2+}$ to which the first and second Cl^- ligands are added to form transplatin (**F**) as represented in the scheme below.

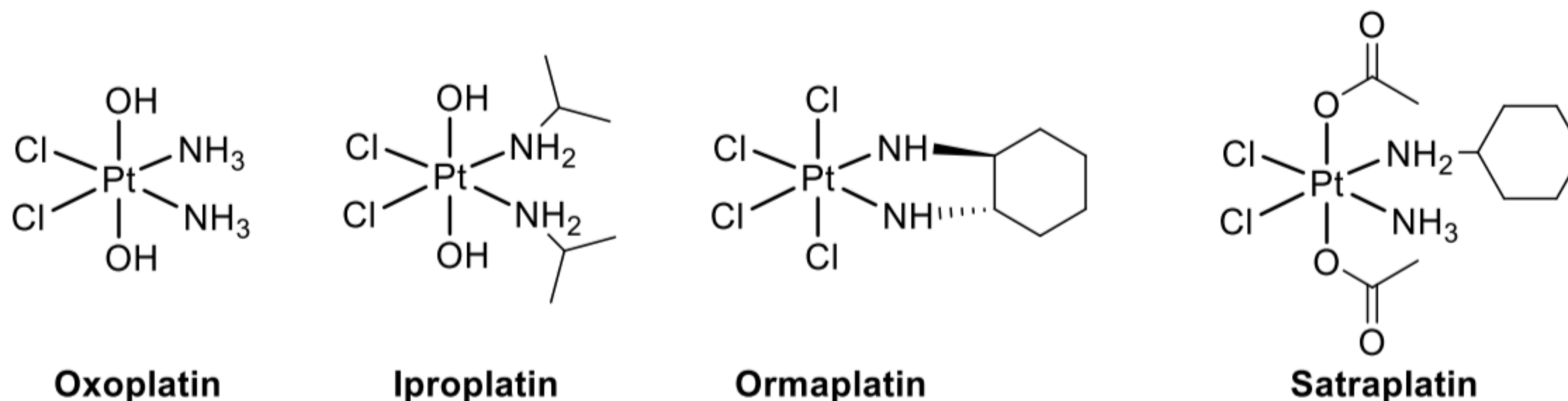


3.6. **Draw** the molecular structures of **E** and **F**.

The most important classes of antitumor agents, cisplatin, carboplatin, and oxaliplatin as platinum(II) diamines are widely used in chemotherapy to treat a wide variety of cancers.



However, the therapeutic index of these agents is relatively narrow; their use is often plagued with severe toxicity and the development of resistance, which leads to disease progression. Recently, oxoplatin, iproplatin, ormaplatin and satraplatin are Pt complexes that have been used clinically (oxoplatin) or in clinical trials.



3.7. All complexes have the same geometry and oxidation number for the Pt central atom.

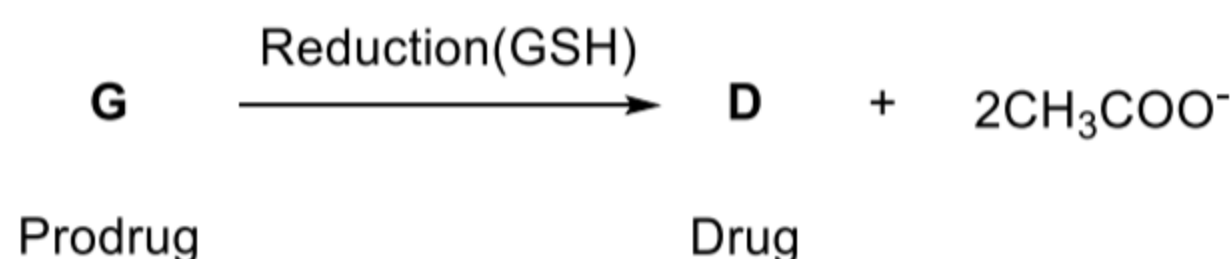
Write the oxidation state of Pt and geometry of the complexes.

3.8. Which Pt complex, cisplatin or satraplatin, is kinetically more inert for substitution reactions?

3.9 Oxaplatin is an isomer of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2(\text{OH})_2]$ complex. **Draw** all stereoisomers and **indicate** the chiral one(s).

Platinum complexes (oxoplatin, iproplatin, ormaplatin, and satraplatin) can be considered prodrugs that are primarily intracellularly activated by biological reducing agents such as thiols, ascorbic acid, and glutathione (GSH) to kill cancer cells.

In a study, for example, the reduction of *cis,trans,cis*- $[\text{PtCl}_2(\text{OCOCH}_3)_2(\text{NH}_3)_2]$ (**G**, prodrug), which has a similar structure to satraplatin, by aqueous extract of cancer cells (A2780, A2780cisR, and HT-29) yields cisplatin (**D**, drug) and free acetate ion as given below.



3.10. **Draw** the molecular structure of **G**.

3.11. **Sketch** the d-orbital splitting of the metal ion in **G** and **write** the electronic configuration.

3.12. **Decide** whether **G** is paramagnetic or diamagnetic.

3.13. The complex **G** crystallizes into a monoclinic crystal system of parameters: the lengths of the unit cell: $a = 14.9973$, $b = 8.57220$, $c = 11.1352$ Å, the β angle in the unit cell = 126.7690° , the number of the molecules in the unit cell (Z) = 4, $M = 436.16$ g/mol (the complex has one water molecule in the crystal structure).

Calculate the density (ρ) of the complex. *Hint*: the volume of a monoclinic crystal unit cell is $V = a \times b \times c \times \sin \beta$